1. Shared Hosting vs VPS vs Cloud Hosting

Comparison

Feature	Shared Hosting	VPS	Cloud Hosting
Cost	Very Low	Medium	Pay-as-you-go
Performance	Low (shared resources)	Good	High
Scalability	Poor	Moderate	Excellent
Control	Minimal	Root access	Full control
Reliability	Low	Medium	High

Recommendation

For growing traffic → **Cloud Hosting**

- Auto scaling
- No downtime during traffic spikes
- Better performance than shared/VPS

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2. E-commerce Website During Peak Traffic

Architecture Design

Components:

- Load Balancer (distributes traffic)
- Auto Scaling Servers
- CDN (for static content)
- Database Replication
- Cache (Redis)

Flow:

User → CDN → Load Balancer → Multiple Servers → Database Cluster

Key Features:

- Horizontal scaling
- Failover support
- Zero downtime

👉 Used by companies like Amazon during sales

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3. Website Migration (Shared → Cloud)

Migration Plan

1. Backup
    - Full website files
    - Database dump
2. Setup Cloud Server
    - Install environment (Apache/Nginx, MySQL)
3. Deploy Website
    - Upload files
    - Import database
4. Testing
    - Check performance
    - Fix errors
5. DNS Switch
    - Update domain to new server
6. Downtime Minimization
    - Use low TTL
    - Perform migration during low traffic

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4. Cost Optimization in Hosting

Strategies

- Use **Auto Scaling**
- Turn off unused instances
- Use **Reserved Instances**
- Use CDN to reduce server load
- Optimize database queries
- Use serverless (Lambda)

👉 Example: Netflix optimizes cost with cloud scaling

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5. Managed vs Unmanaged Hosting

Comparison

Feature	Managed	Unmanaged
Technical Work	Provider handles	User handles
Cost	Higher	Lower
Ease	Easy	Complex

Recommendation

For startup with no expertise → **Managed Hosting**

👉 Providers like Bluehost offer managed services

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6. Cloud Hosting Architecture Design

Architecture

- Load Balancer
- Application Servers (Auto Scaling)
- Database (Primary + Replica)
- CDN
- Storage (S3)

Flow:

User → CDN → Load Balancer → App Servers → DB

👉 Used by Netflix

Benefits:

- High availability
- Global scalability
- Fault tolerance

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7. CDN and Website Performance

Problem:

Slow for global users

Solution:

Use CDN

How it works:

- Stores cached content in global servers
- Serves user from nearest location

👉 Example: Cloudflare

Benefits:

- Faster loading
- Reduced latency
- Lower server load

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8. Security in Web Hosting

Security Audit Checklist

- Enable HTTPS (SSL)
- Use Firewall (WAF)
- Strong passwords
- Regular updates
- Backup system
- DDoS protection

Improvements:

- Use IAM roles
- Encrypt data
- Monitor logs

👉 Example: Equifax Data Breach

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9. SaaS Application Hosting

SaaS Architecture

- Multi-tenant system
- Load balancer
- Scalable backend
- Central database

Features:

- High availability
- User isolation

- Subscription model

👉 Example: Google SaaS services

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10. Website Failure (Downtime Case)

Root Cause Analysis

Common issue:

- DNS misconfiguration
- Server overload
- Deployment error

👉 Example: Facebook Outage 2021

Prevention:

- Monitoring tools
- Backup DNS
- CI/CD testing
- Rollback mechanisms

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11. Enterprise/Government Cloud Adoption

Advantages:

- Scalability
- Cost efficiency
- Remote access

Challenges:

- Data security
- Compliance issues
- Vendor lock-in

👉 Example: Government of India cloud initiatives

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12. Local Business Website Hosting

Recommendation

👉 **Shared Hosting** (best for small business)

Why:

- Low cost
- Easy to use
- Enough for low traffic

👉 Providers like Hostinger

✅ **Final Summary**

- Small sites → Shared Hosting
- Growing sites → VPS
- High traffic / scalable → Cloud
- No technical team → Managed Hosting
- Global users → CDN
- Security → Must be priority